A Text-Based Geopolitical Alignment Index from Foreign Ministry Press Releases

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*The views expressed in this paper are those of the author(s) and do not necessarily reflect the views of the Federal Reserve System, the Board of Governors, or the regional Federal Reserve Banks

The Problem:

- We're addressing a **gap in existing measurements of geopolitical alignment** , the Bailey et al 2017 method remains widely used; but it relies on voting patterns in the UNGA to create its index. Our research builds on this by creating an index based on press releases from the foreign ministries of different countries.
- We've built two Geopolitical Indices:
  - **Usage-Weighted Alignment Scores:** uses proximity-aware sentiment attribution and usage weighting to quantify how positively or negatively an origin country publicly characterizes a target country.
  - **LLM-Based Geopolitical Alignment Index:** where the LLM identifies the following objects in texts it recieves:

a. The Text
b. Origin Country
c. Target country

AI’s Role:

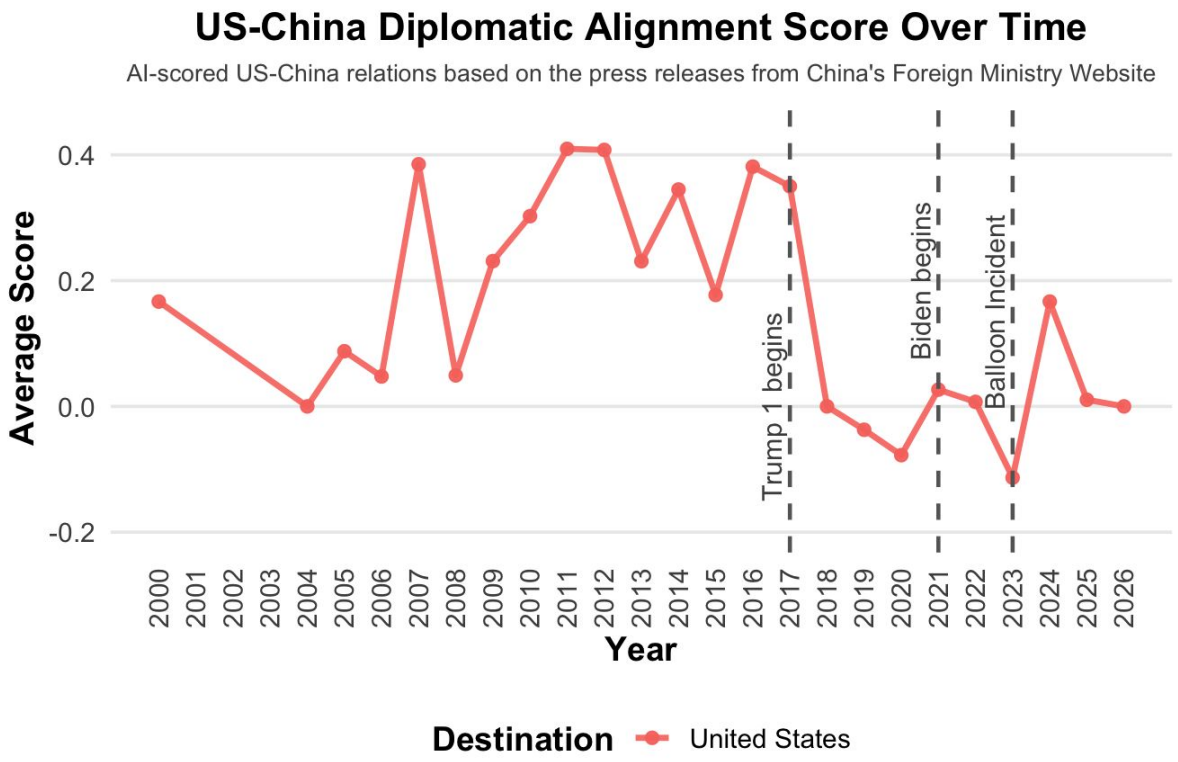
- We’ve used AI to **scrape government websites** for text, write code on the usage weighted alignment method, create an LLM based alignment measure, and use ChatGPT 5.2 or 5.0.nano through its API to assign scores, as it understands **context** .

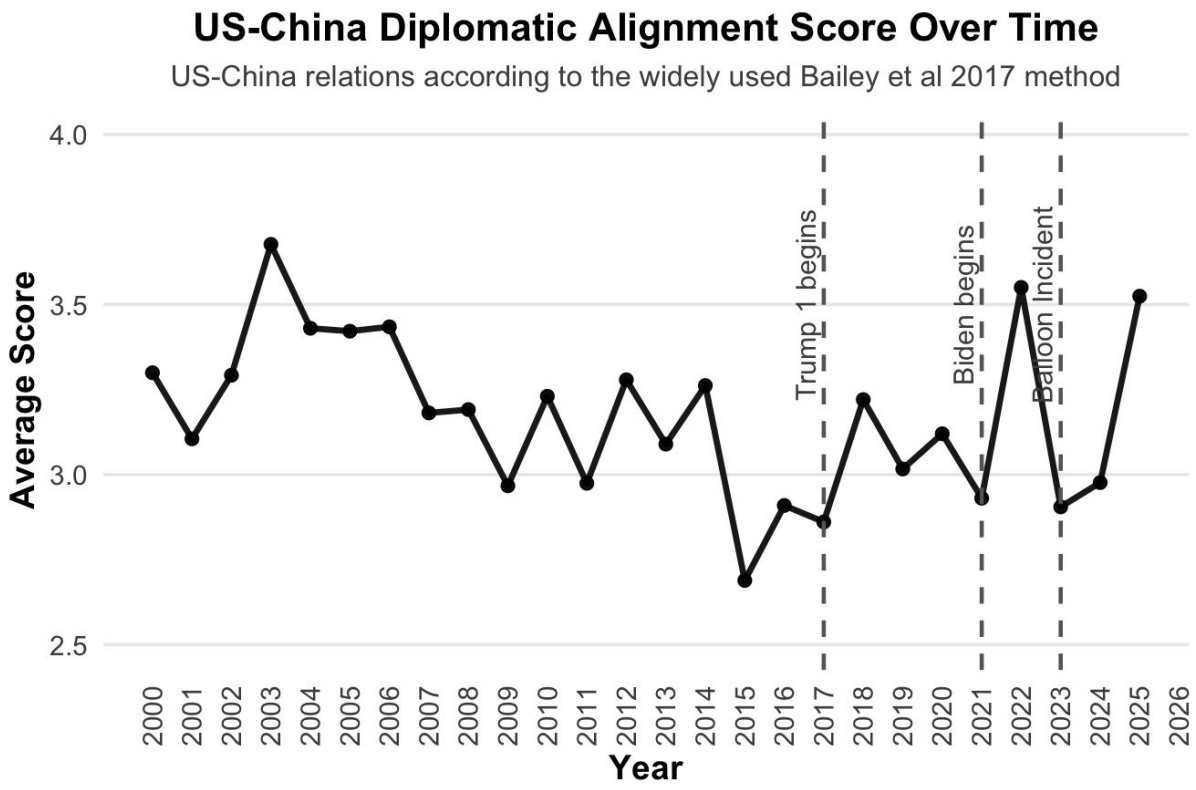
- AI captures **subtle diplomatic tone** as foreign ministry texts rarely use overt language, and a ChatGPT-style model is able to interpret this more aptly.

- Finally, it can focus on the **specific bilateral relationship** within an extensive document. If a statement mentions several countries, the model can isolate the language about China–France or India–Liberia, instead of averaging the whole document’s tone.

Results and Impact:

- Using the country pair of US - China, our **LLM based mode** l has delivered the following results, with results **differing notably** from the current UN data based model.





Appendix

- This index has practical applications across sectors, and its reliance on public data and automated natural language processing further ensures its **scalability** across **countries, languages and multiple time frames** in any given context.
- As an example, using Canadian press releases from 2024 our usage weighted method has shown Canada to align most closely with the following 5 countries:

origin	country	iso3c	n_docs	pos_total	neg_total	net_total
CAN	United States	USA	17	73.44	25.76	42.55
CAN	Ukraine	UKR	32	392.03	298.03	34.39
CAN	Venezuela	VEN	5	26.27	0	26.27
CAN	Hong Kong SAR China	HKG	4	22.28	0	22.28
CAN	Mexico	MEX	6	19.07	0	19.07